

Literature Status: Conditional Bases with Property (A)

Automatic Math Discovery Run `fa.banach.001`

June 14, 2026

Original Problem

The source paper is Fernando Albiac, José L. Ansorena, and Vladimir Temlyakov, *Twenty-five years of greedy bases*, arXiv:2405.20939. In Section 4, “Isometric almost greediness”, PDF page 11, Problem 6 asks:

Does Property (A) imply unconditionality?

The same page notes that it was also unknown whether 1-suppression quasi-greediness implies unconditionality.

Supporting Answer Source

The supporting paper is Fernando Albiac, José L. Ansorena, Miguel Berasategui, and Pablo M. Berná, *Conditional bases with Property (A)*, arXiv:2409.04883. Its introduction cites the source paper as AAT2024 and says that Question G was explicitly raised in AAT2024, Problem 6. On PDF page 4 it states that the paper answers Question G negatively.

Answer

The 2405.20939 question is already answered negatively in the later literature. Theorem 3.3 of arXiv:2409.04883 constructs Schauder bases with Property (A) whose unconditionality parameters are unbounded when the chosen symmetric space properly couples with the weight. Corollary 3.7 then states explicitly that there exist Banach spaces with a bidemocratic Schauder basis which fails to be unconditional and yet has Property (A). Hence Property (A) does not imply unconditionality.

The supporting paper also addresses the adjacent question mentioned after Problem 6 in the source paper: on PDF page 17 it states that, since a basis with Property (A) is 1-almost greedy and therefore 1-suppression quasi-greedy, Theorem 3.3 answers negatively whether 1-suppression quasi-greediness implies unconditionality. It then gives a reflexive version with a conditional, bimonotone, isometrically bidemocratic 1-suppression quasi-greedy basis.

Verification Notes

Local source text checks:

- `data/parsed/arxiv_sources/2405.20939/source.tex`, lines 487–493, contains Problem 6 and the nearby 1-suppression quasi-greedy question.

- `data/parsed/arxiv_sources/2409.04883/source.tex`, lines 202–214, states Question G and says it is raised in AAT2024 Problem 6 and answered negatively.
- `data/parsed/arxiv_sources/2409.04883/source.tex`, lines 587–599 and 707–715, contain Theorem 3.3 and Corollary 3.7.
- `data/parsed/arxiv_sources/2409.04883/source.tex`, lines 776–787, records the negative answer to the 1-suppression quasi-greedy variant.

Search Bounds

Before making this packet, the run registry, solution index, attempt index, proof-gap index, existing solution folders, and existing attempt folders were searched for arXiv:2409.04883, the title phrase “Conditional bases with Property (A)”, and the Property (A) unconditionality question. No existing run packet or ledger row was found. Web search on June 14, 2026 found the arXiv record for arXiv:2409.04883, whose abstract says the paper builds Schauder bases with Property (A) that fail to be unconditional.

Limitations

This is a literature-status packet, not a new proof or counterexample produced by the run. It records that the extracted open problem from arXiv:2405.20939 was answered by a later paper. It does not claim any resolution of the new questions raised at the end of arXiv:2409.04883.

Human Review Recommendation

Check that AAT2024 in arXiv:2409.04883 is exactly arXiv:2405.20939 and that the cited Problem 6 is the same Property (A) unconditionality problem recorded in the source paper. The identification is explicit, so this should be a straightforward literature-already-answered entry.